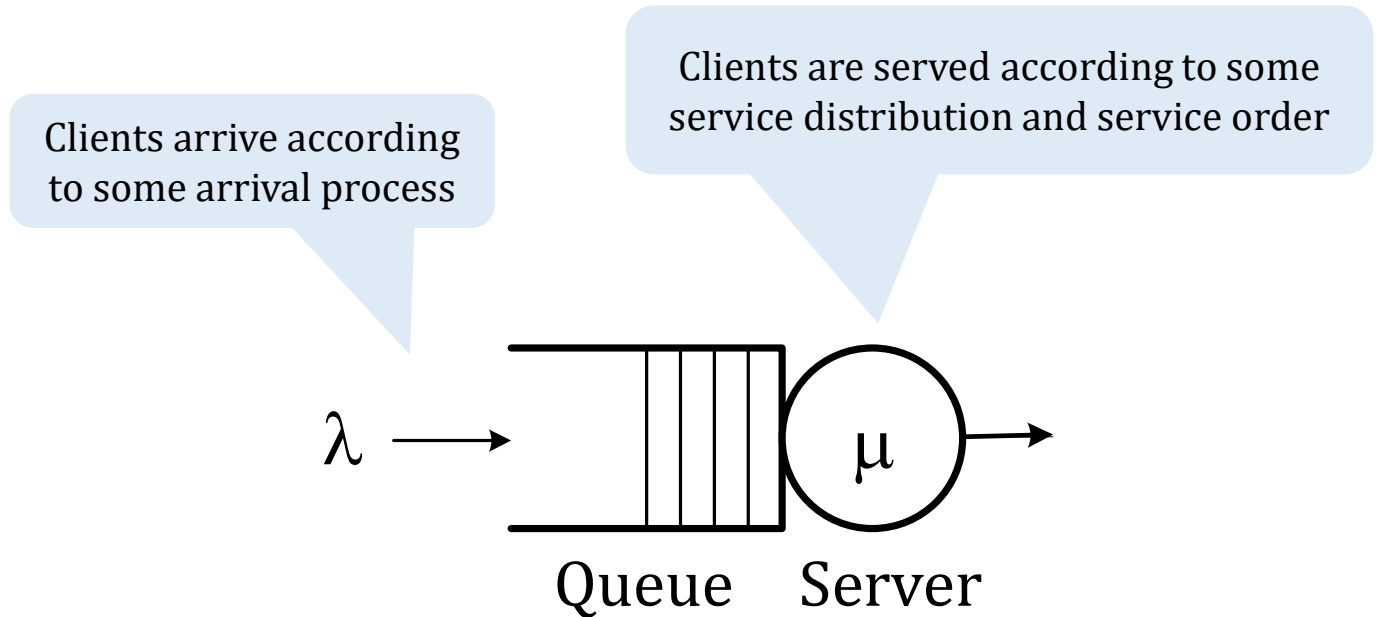




TÉCNICO
LISBOA

Queuing systems and applications

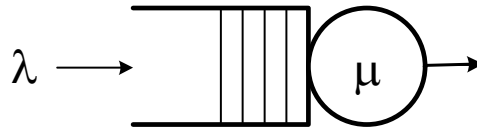
Single-server queuing system



Many types of clients: packets, calls, CPU jobs, people, cars, ...

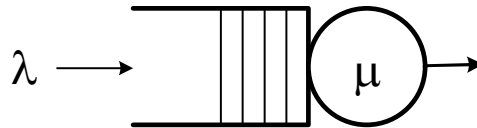
Single-server queuing system

- Single-server queuing system parameters
 - Service order – FCFS, strict priority, Processor-Sharing, ...
 - Average arrival rate – λ clients/sec, jobs/sec, ...
 - Mean interarrival time – $1/\lambda$ sec
 - Service size – S service units/client, bits/packet, CPU cycles/job
 - Average service rate – μ clients/sec, jobs/sec, ...
 - Mean service time – $1/\mu$ sec



Single-server queuing system

- Performance metrics
 - Average delay in system – W
 - Average delay in queue – W_Q
 - Average number of clients in system – L
 - Average number of clients in queue – L_Q
 - Server utilization – ρ
 - Throughput – X
 - Blocking probability – P_B



Service size and speed

- Service can be expressed directly as service time required by client
- Service can be expressed as **absolute server speed + service size of client**
 - Client is packet, server is transmission link, server speed is link capacity in bits/s, service size is packet size in bits/packet
 - Client is CPU job, server is CPU, server speed in CPU speed in cycles/s, service size is job service requirement in cycles/job

$$\text{service time} = \frac{\text{service size (service units / client)}}{\text{server speed (service units / time unit)}}$$

Kendall's notation



A/S/c/K/N/D

M / M / 1

M / M / K / K

M / 6 / 1

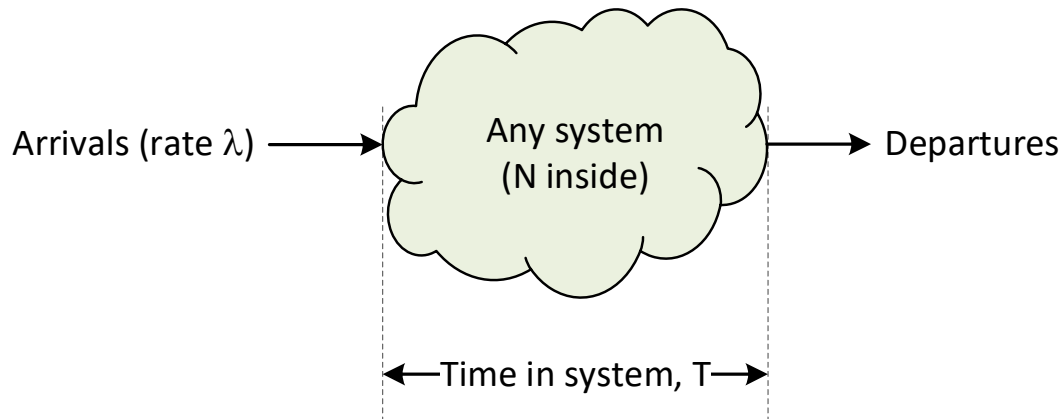
- A – Arrival process
- S – Service time distribution
- c – Number of servers
- K – System capacity
- N – Size of the population from which the customers come
- D – Service discipline

Little's Law

- For any ergodic open system

$$L = \lambda W$$

where $L = E(N)$ is the average number of clients in the system, λ is the average arrival rate into the system, and $W = E(T)$ is average delay in system.

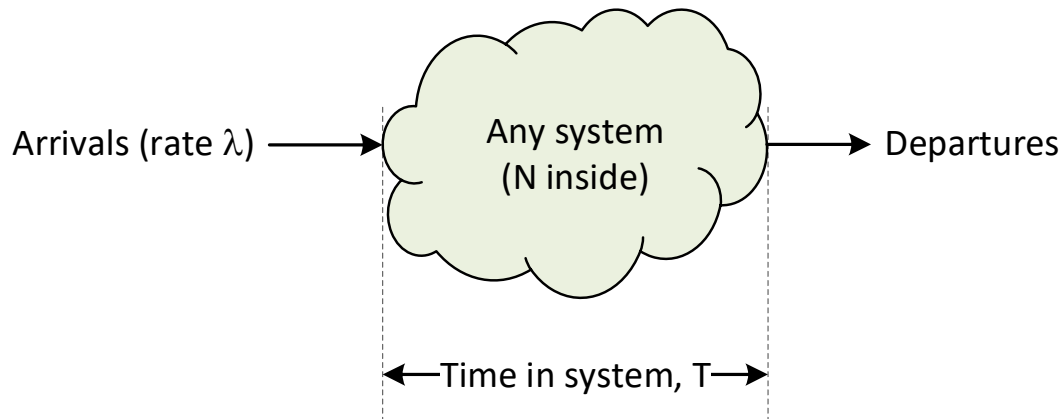


Little's Law

- L , λ and W can also be seen as time averages, since the system is ergodic.
- Let δ be the departure rate. It can be show that Little's law holds if the time averages L , W , λ and δ exist, and

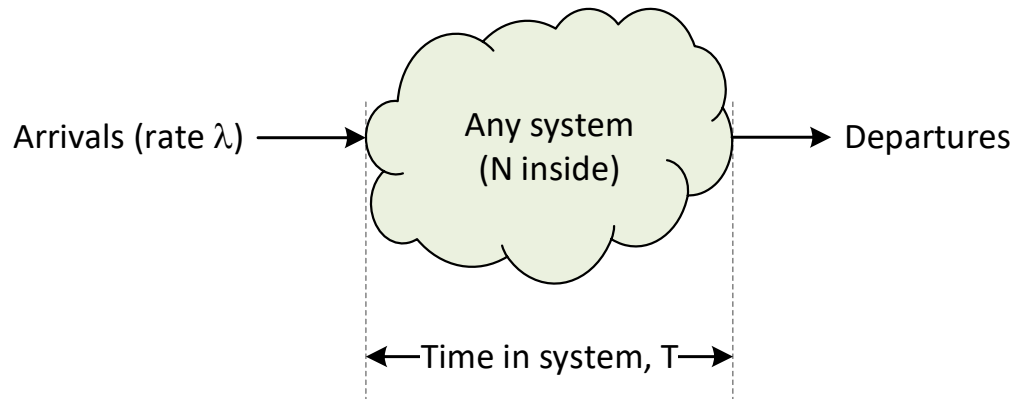
$$\lambda = \delta$$

i.e. the system does not creates or deletes clients.



Little's Law

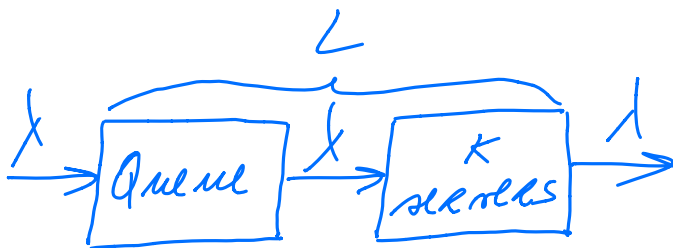
- Little's law expresses the intuitive idea that congested systems (high L) have high delays (high W):
 - In a rainy day, the traffic delay is higher (higher W) and the streets are more congested (higher L), for the same traffic (equal λ).
 - A fast food restaurant (lower W) needs a smaller room (lower L) than a normal restaurant, for the same customer arrival rate (equal λ).



Little's Law

$$L = \lambda W$$

- Consider a system with K servers and capacity for at most $L \geq K$ customers (waiting in the queue or being served). The system is always full: suppose that there are initially L customers and that, as soon as one customer leaves the system, it is immediately replaced by another one that arrives. Assume also that the average service time per customer is $\bar{X}$. What is the average delay in system, as a function of L , $\bar{X}$, and K ?



Servers: $K = \lambda \bar{X} \quad \lambda = \frac{K}{\bar{X}}$

Total: $W = \frac{L}{\lambda} = \frac{L \bar{X}}{K}$

Utilization and throughput

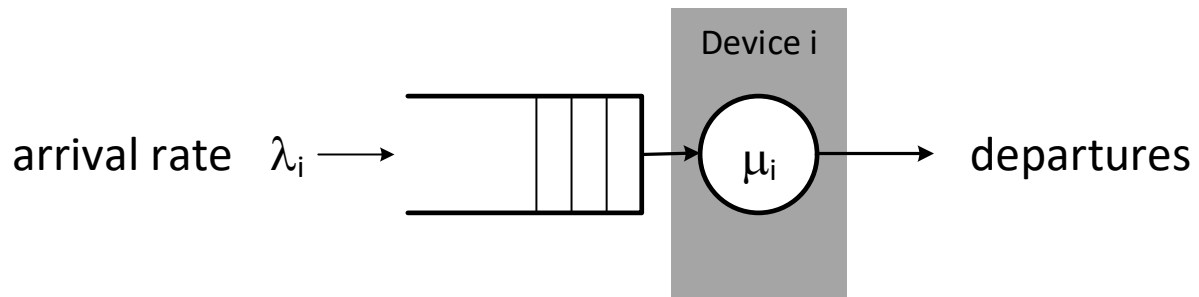
- Device **utilization** (ρ_i): fraction of time device i is busy
- Device **throughput** (X_i): rate of completions at device i (e.g. clients/sec)
- If τ is a long observation period, B is the total time the device is busy, and C is the total number of completed jobs:

$$\rho_i = \frac{B}{\tau} \qquad X_i = \frac{C}{\tau}$$

Utilization law

- A corollary of Little's law
- Consider a single device i with average arrival rate λ_i clients/sec and average service rate μ_i clients/sec, where $\lambda_i < \mu_i$. Let ρ_i denote the long-run fraction of time that the device is busy. Then:

$$\rho_i = \frac{\lambda_i}{\mu_i}$$



Utilization law: $\rho = \frac{\lambda}{\mu}$

ERGODICITY $\Rightarrow \rho$ $\left\{ \begin{array}{l} \rightarrow \text{long-run fraction of} \\ \text{time device is busy} \\ \rightarrow \text{ensemble average} \\ \text{that device busy} \end{array} \right.$

Device is busy with prob. ρ $\rightarrow 1 \text{ client}$
" " idle " " $(1-\rho)$
 $\hookrightarrow 0 \text{ clients}$

Expected number of clients in
device: $\rho \times 1 + (1-\rho) \times 0 = \rho$

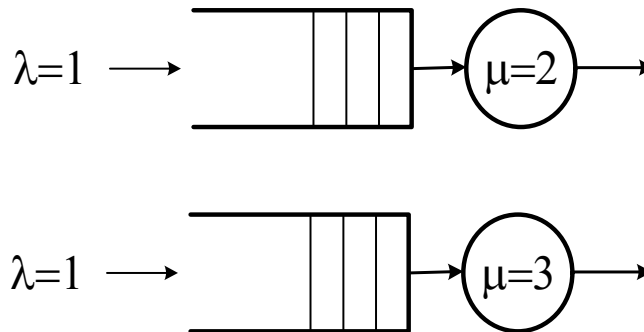
$$\rho = \lambda W = \frac{\lambda}{\mu}$$

$\hookrightarrow$ Little's law

$$X = \frac{n^{\circ} \text{ of completions}}{t}$$

Throughput

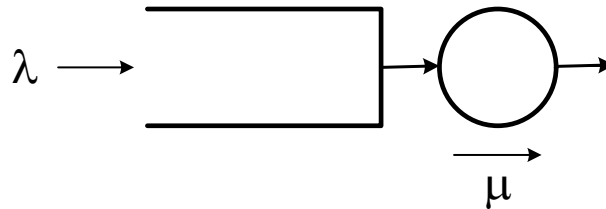
- The throughput of a queuing system is $X = \lambda$ since the completion rate is bounded by the arrival rate, i.e. **rate in = rate out**.



Systems with same throughput (but different response time, i.e. delay)

M/M/1 system

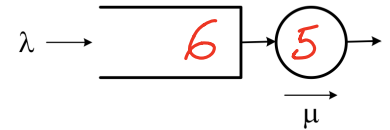
- Queuing system where:
 - There is a single server
 - Customers arrive according to a Poisson process with rate λ
 - The service time is exponentially distributed with mean $1/\mu$
 - The arrival process is independent of the service time
 - The system accommodates an infinite number of customers



Limiting state probabilities of M/M/1

- Birth-death process with $\lambda_n = \lambda$ and $\mu_n = \mu$.

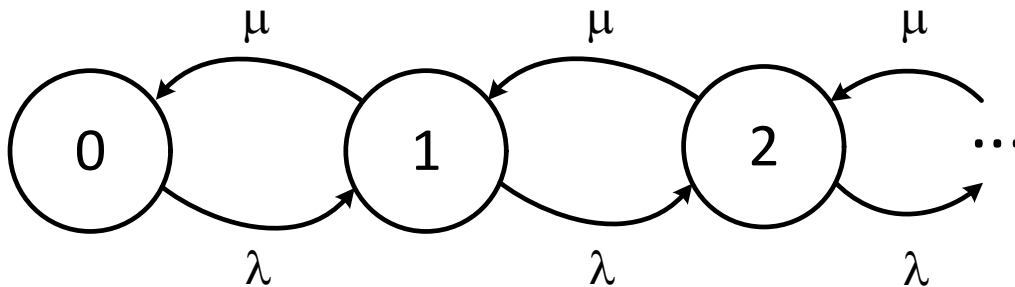
$$\pi_n = \frac{(\lambda/\mu)^n}{1 + \sum_{i=1}^{\infty} (\lambda/\mu)^i}$$



- If $\lambda/\mu < 1$

Stability condition of M/M/1

$$\pi_n = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n, \quad n \geq 0$$



$$\pi_n = \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1} \cdot \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1}}, \quad n \geq 1$$

$$\lambda_n = \lambda, \quad \mu_n = \mu$$

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \cdot \frac{1}{1 + \sum_{n=1}^{\infty} \left(\frac{\lambda}{\mu}\right)^n}$$

$$= \frac{\left(\lambda/\mu\right)^n}{\sum_{n=0}^{\infty} \left(\lambda/\mu\right)^n} = \frac{\left(\lambda/\mu\right)^n}{\frac{1}{1 - \lambda/\mu}}$$

$\xrightarrow{\lambda/\mu < 1}$

$$= \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right), \quad n \geq 0$$

M/M/1 formulas

- Average number of customers in system:

$$L = \sum_{n=0}^{\infty} n\pi_n = \frac{\lambda}{\mu - \lambda}$$

- Average delay in system:

$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}$$

- Average delay in queue:

$$W_Q = W - E(S) = W - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)}$$

M/M/1 formulas

- Average number of customers in queue:

$$L_Q = \lambda W_Q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

- Average number of customers in server (server utilization):

$$L_S = \rho = \frac{\lambda}{\mu}$$

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right), \quad n \geq 0$$

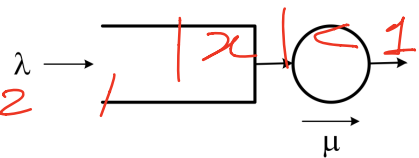
$$\begin{matrix} 0 & 1 & 2 & \dots \\ \pi_0 & \pi_1 & \pi_2 & \dots \end{matrix}$$

$$L = \sum_{n=0}^{\infty} n \cdot \pi_n = \left(1 - \frac{\lambda}{\mu}\right) \sum_{n=0}^{\infty} n \left(\frac{\lambda}{\mu}\right)^n$$

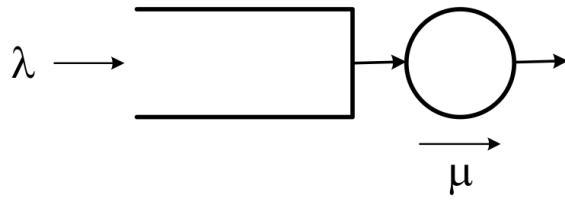
$$= \left(1 - \frac{\lambda}{\mu}\right) \cdot \frac{\lambda/\mu}{(1 - \lambda/\mu)^2} = \frac{\lambda/\mu}{(1 - \lambda/\mu)}$$

$$= \frac{\lambda}{\mu - \lambda}$$

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2}$$



$$L = \frac{\lambda}{\mu - \lambda}$$



$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}$$

$$W_q = W - W_s = W - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$L_q = \lambda W_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

$$L_s = \rho = \frac{\lambda}{\mu}$$

M/M/1 increasing the arrival and service rates proportionally

- If the arrival rate λ and service rate μ are increased by a factor k , it results:
 - Server utilization: $\rho_{new} = \rho_{old}$
 - Throughput: $X_{new} = k\lambda_{old}$
 - Average number of clients in system: $L_{new} = L_{old}$
 - Average delay of clients in system: $W_{new} = \frac{W_{old}}{k}$
- Interesting: the delay drops indefinitely with k !

$$\lambda_N = K \lambda_0 \quad \mu_N = K \mu_0$$

$$\rho_N = \frac{\lambda_N}{\mu_N} = \frac{K \lambda_0}{K \mu_0} = \rho_0$$

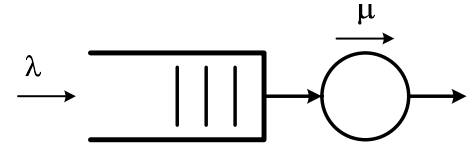
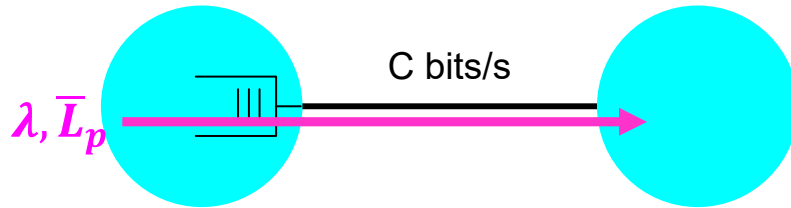
$$L_N = \frac{\lambda_N}{\mu_N - \lambda_N} = \frac{K \lambda_0}{K \mu_0 - K \lambda_0} = \frac{\lambda_0}{\mu_0 - \lambda_0} = L_0$$

$$W_N = \frac{1}{\mu_N - \lambda_N} = \frac{1}{K \mu_0 - K \lambda_0} = \frac{W_0}{K}$$

$$X_N = \lambda_N = K \lambda_0 = K X_0$$

Client is packet, Queue is buffer,
Server is transmission link, packets have size

Point-to-point links and M/M/1

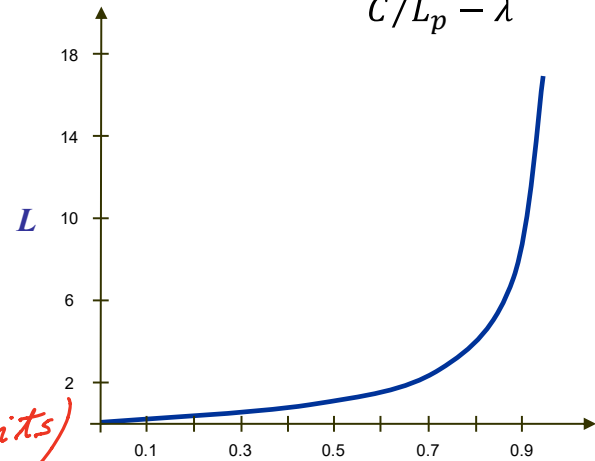


$$W = \frac{1}{C/\bar{L}_p - \lambda}$$

$$L = \frac{\lambda}{C/\bar{L}_p - \lambda}$$

Point-to-point link = M/M/1:

- ❖ Packet switching
- ❖ Poisson arrivals at rate λ
- ❖ Exponentially distributed packet lengths
 $\Rightarrow$ Service rate $\mu = C/\bar{L}_p$
- ❖ Independence of arrivals and lengths
- ❖ Infinite capacity buffer



Time tx packet = $\frac{\text{packet size (bits)}}{\text{capacity (bits/s)}}$

$\lambda/\mu \leftarrow$ link utilization

Point-to-point links and M/M/1

- The access link of an ISP has a capacity of $C = 155$ Mbits/s. The clients of the ISP that use this access link generate collectively a Poisson packet flow with rate $\lambda = 15.5$ Kpackets/s and exponentially distributed packets sizes with average $\bar{L}_p = 8$ Kbits. Assume that the buffer capacity of the access link is very large compared with the average buffer occupancy and that the packet arrival process is independent from the packet sizes. Determine the average packet delay imposed by the access link and the link utilization.

M/M/1

$$W = \frac{1}{\mu - \lambda} = \frac{1}{19.375K - 15.5K} = 258 \mu s$$

$$\lambda = 15.5 \text{ Kpackets/s}$$

$$\mu = \frac{C}{\bar{L}_p} = \frac{155M}{8K} = 19.375 \text{ Kpackets/s}$$

$$\rho = \frac{\lambda}{\mu} = 0.8$$

Point-to-point links and M/M/1

- Suppose you are a Networking Engineer at an ISP. You were asked the following question by your boss (which knows nothing about network dimensioning): if the traffic of customers from one access network doubles, how much additional capacity should we buy, to keep the same Quality-of-Service (expressed in terms of average packet delay)? (You have checked that the access network is suitably modelled by the M/M/1 queuing system)
- The following relationship holds when the arrival rate is increased by a factor of k and the average delay is kept the same:

$$\frac{C_{new}}{C_{old}} = \frac{\mu_{new}}{\mu_{old}} = 1 + (k - 1)\rho_{old}$$

$$\lambda_N = K \lambda_0$$

$$W_N = W_0$$

$$W_N = \frac{1}{\mu_N - \lambda_N} = \frac{1}{\mu_N - K \lambda_0} = \frac{1}{\mu_0 - \lambda_0}$$

$$\mu_N - K \lambda_0 = \mu_0 - \lambda_0$$

$$\frac{\mu_N}{\mu_0} = K \left(\frac{\lambda_0}{\mu_0} \right) + 1 - \left(\frac{\lambda_0}{\mu_0} \right) = \rho_0$$

$$= 1 + (K-1) \rho_0$$

$$\rho_0 < 1 \Rightarrow (K-1) \rho_0 < K-1$$

$$\Rightarrow \frac{\mu_N}{\mu_0} = \frac{C_N}{C_0} < K$$

Point-to-point links and M/M/1

- The marketing department of the ISP estimates that the traffic generated at the access link will triplicate in the near future. What capacity should be installed in order to maintain the same quality-of-service (expressed in terms of average packet delay)?

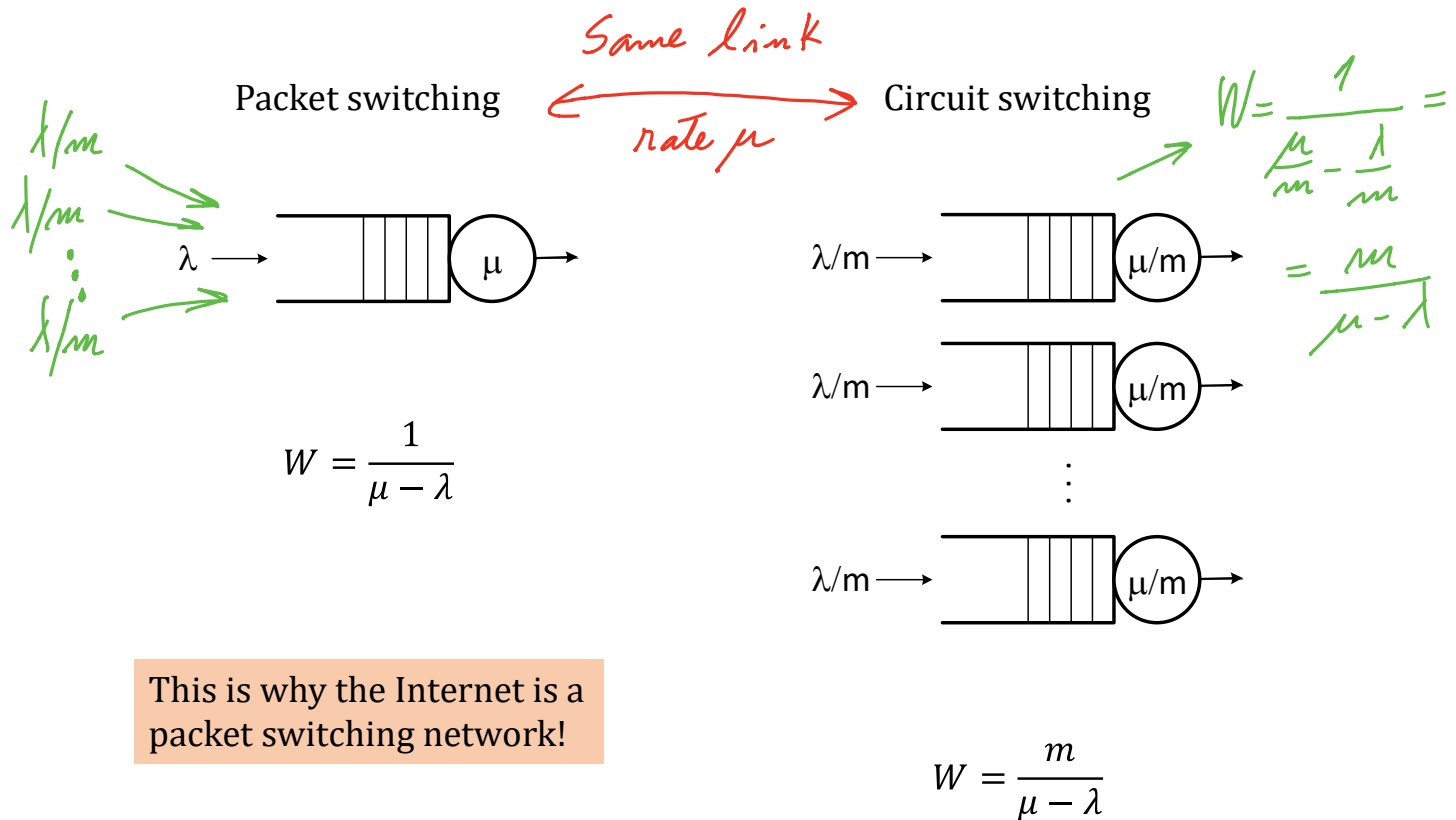
$$\lambda_N = 3\lambda_0 \Rightarrow K = 3$$

$$\rho_0 = 0.8$$

$$\frac{\mu_N}{\mu_0} = \frac{C_N}{C_0} = 1 + (K-1)\rho_0 = 1 + 2\rho_0 = 2.6$$

$$C_N = 2.6 \times 155 \text{ M} = 403 \text{ Mb/s}$$

Packet switching versus circuit switching

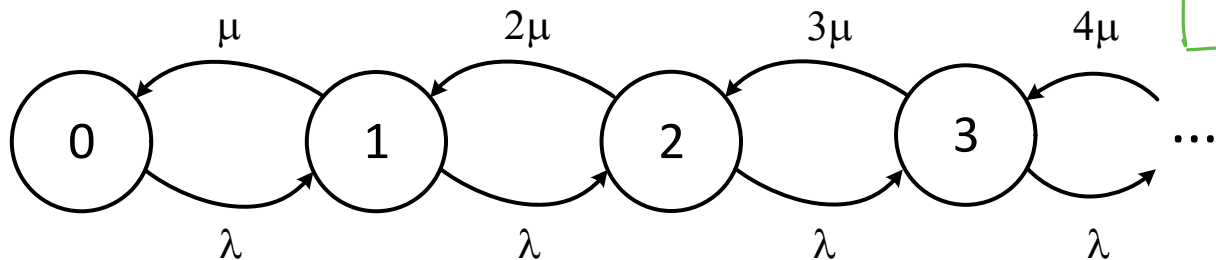
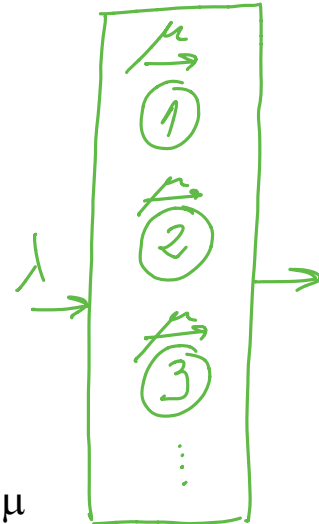


M/M/ ∞ system

$$T_i \sim \text{Exp}(\mu_i)$$
$$\min(T_1, T_2, \dots) \sim \text{Exp}(\mu_1 + \mu_2 + \dots)$$

- Queuing system where:

- There are an infinite number of servers
- Customers arrive according to a Poisson process with rate λ
- The service time is exponentially distributed with mean $1/\mu$
- The arrival process is independent of the service time
- The system accommodates an infinite number of customers





- * 2 clients being served
- * exponentially distributed service times at rate μ
- * $T_1 \equiv$ time until client 1 dies
- $T_2 \equiv$ " " " 2 "
- * $\min(T_1, T_2)$ time until there is transition state 1
- * $\min(T_1, T_2) \sim \text{Exp}(2\mu)$

M/M/ ∞ formulas

- Balance equations:

$$\lambda\pi_{n-1} = n\mu\pi_n, \quad n = 1, 2, \dots$$

- The limiting probability of n clients in the system is Poisson distributed with parameter λ/μ :

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \frac{e^{-\lambda/\mu}}{n!}$$

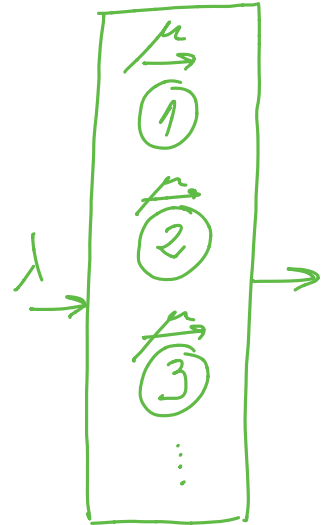
M/M/ ∞ formulas

- Average number of clients in system:

$$L = \frac{\lambda}{\mu}$$

- Average delay in system:

$$W = \frac{1}{\mu}$$



PASTA

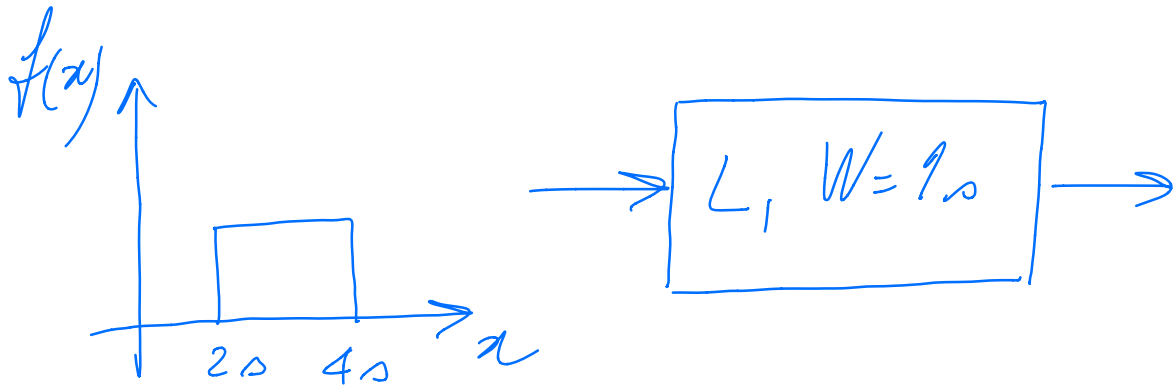
- What is the state of a system from the perspective of an arrival?
- For an ergodic continuous-time system:
 - Let π_n be the limiting probability that **there are n clients in the system** (or the long-run fraction of time that there are n clients in the system)
 - Let a_n be the limiting probability that **an arrival sees n clients in the system** (or the long-run proportion of arrivals that sees n clients)
- a_n is not necessarily equal to π_n
- **PASTA “Poisson Arrivals See Time Averages”**: if the arrival process is Poisson, then

$$a_n = \pi_n$$

- NOTE: interarrival times and service times must be independent

PASTA

- Counter examples:
 - Consider a system where customers arrive with uniformly distributed inter-arrival times between 2 and 4 seconds and the service time is 1 second for each customer. Thus, customers arriving at the system always find the system empty (i.e. $a_0 = 1$, $a_n = 0$ for $n > 0$). On the other hand, the average number of clients in system (as seen by an external observer, that looks at the system at random times) is $1/3$ (and therefore $\pi_0 \neq 1$).
 - Consider a system where the arrival process is Poisson but where the inter-arrival times depend on the services times. Namely, the service time of the n -th customer is half the inter-arrival time between the n th and the $(n + 1)$ -th customer. Thus, customers arriving at the system always find the system empty. However, the average number of customers in the system, as seen by an external observer, is $1/2$.



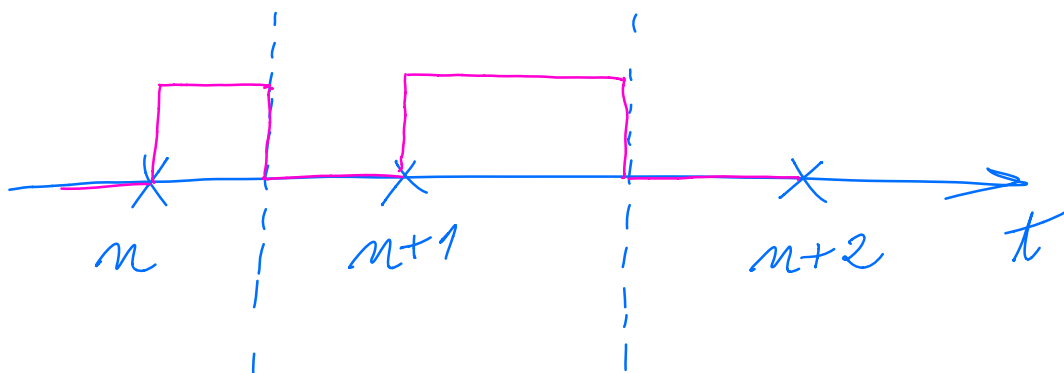
$$\begin{aligned} 1/\lambda &= 3s \\ W &= 1s \end{aligned}$$

$$L = 1/W = \frac{1}{3} \cdot 1 = \frac{1}{3}$$

$$\Rightarrow \pi_0 \neq 1$$

$$a_0 = 1$$

$$a_0 \neq \pi_0$$



$$L = 1/2$$

$$W = (1/\lambda)/2$$

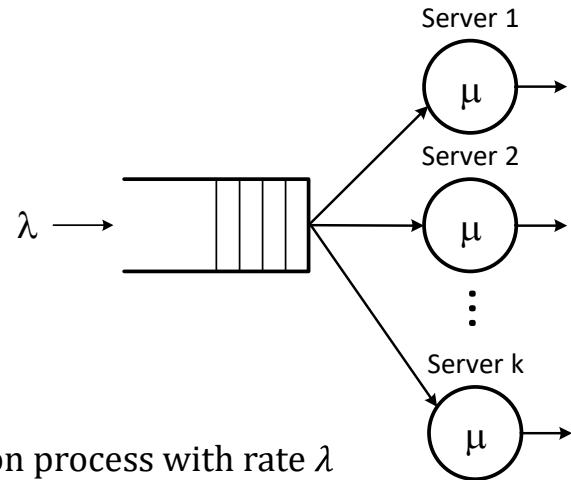
$$L = \lambda W = \cancel{\lambda} \frac{1}{2\cancel{\lambda}} = \frac{1}{2}$$

$$\Rightarrow \pi_0 \neq 1$$

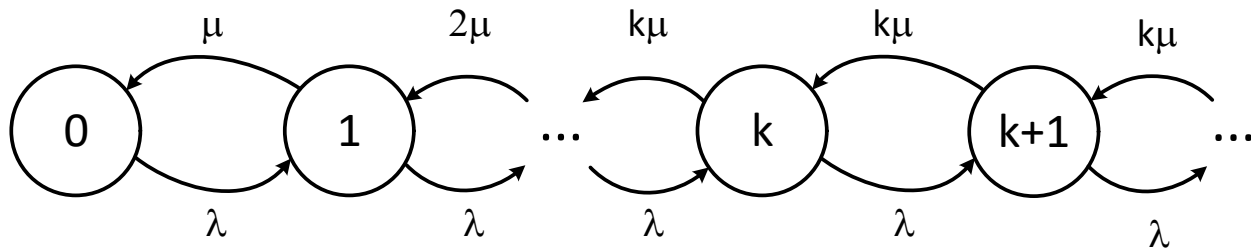
$$a_0 = 1$$

$$a_0 \neq \pi_0$$

M/M/k system



- Queuing system where:
 - There are k servers
 - Customers arrive according to a Poisson process with rate λ
 - The service time is exponentially distributed with mean $1/\mu$
 - The arrival process is independent of the service time
 - The system accommodates an infinite number of customers



M/M/k formulas

stability condition:
 $\rho < 1$

- System utilization:

$$\rho = \frac{\lambda}{k\mu}$$

- Probability that one arrival finds all servers busy (Erlang C formula):

$$P_Q = \sum_{i=k}^{\infty} \pi_i = \frac{(k\rho)^k \pi_0}{k! (1 - \rho)}$$

$$\pi_0 = \left[\left(\sum_{i=0}^{k-1} \frac{(k\rho)^i}{i!} \right) + \frac{(k\rho)^k}{k! (1 - \rho)} \right]^{-1}$$

M/M/k formulas

- Average number of customers in queue:

$$L_Q = P_Q \frac{\rho}{1 - \rho}$$

- Average delay in queue:

$$W_Q = \frac{L_Q}{\lambda} = P_Q \frac{\rho}{\lambda(1 - \rho)}$$

M/M/k formulas

- Average delay in system:

$$W = \frac{1}{\mu} + W_Q = \frac{1}{\mu} + P_Q \frac{\rho}{\lambda(1 - \rho)}$$

- Average number of customers in system:

$$L = \lambda W = \frac{\lambda}{\mu} + P_Q \frac{\rho}{(1 - \rho)}$$

M/M/k

- Expected waiting time for clients who are delayed in the queue:

$$\frac{W_Q}{P_Q} = \frac{1}{k\mu(1 - \rho)}$$

Lesson: High values of system utilization ρ do not imply high delays provided there are sufficiently many servers. Not the same as M/M/1!

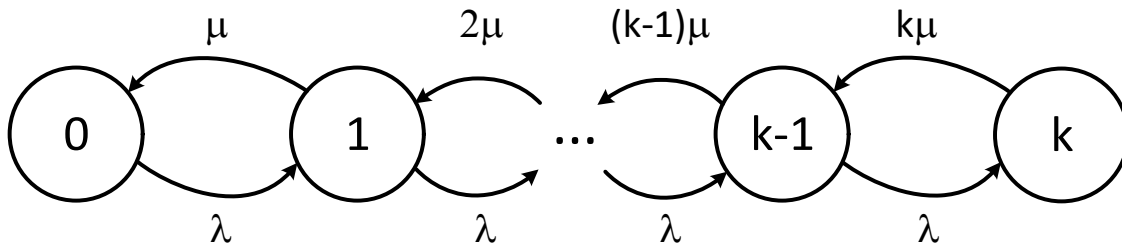
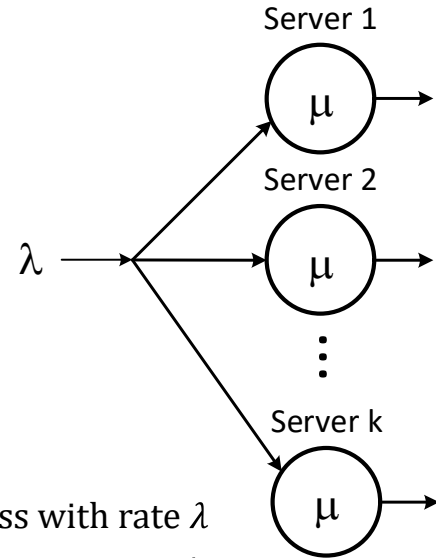
Example:

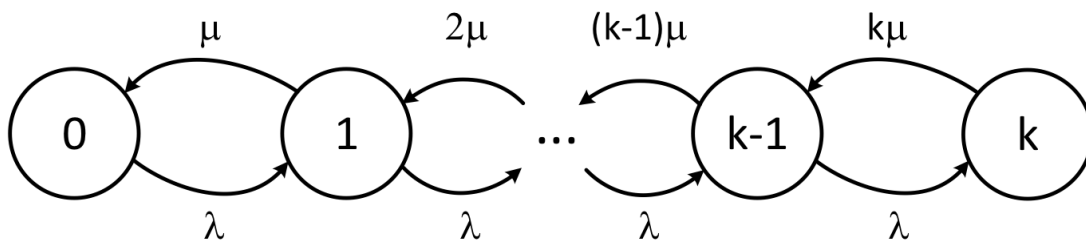
- $\rho = 0.95, k = 5 \Rightarrow \frac{W_Q}{P_Q} = \frac{4}{\mu}$
- $\rho = 0.95, k = 100 \Rightarrow \frac{W_Q}{P_Q} = \frac{1}{5\mu}$

M/M/k/k system

- Queuing system where:

- There are k servers
- Customers arrive according to a Poisson process with rate λ
- The service time is exponentially distributed with mean $1/\mu$
- The arrival process is independent of the service time
- The system can only accommodate k customers





$$\lambda \pi_0 = \mu \pi_1$$

$$\lambda \pi_1 = 2\mu \pi_2$$

$$\lambda \pi_2 = 3\mu \pi_3$$

⋮

$$\lambda \pi_{i-1} = i\mu \pi_i$$

$$\pi_1 = \left(\frac{\lambda}{\mu} \right) \pi_0$$

$$\pi_2 = \left(\frac{\lambda}{2\mu} \right) \pi_1 = \frac{\lambda^2}{2\mu^2} \pi_0$$

$$\pi_3 = \left(\frac{\lambda}{3\mu} \right) \pi_2 = \frac{\lambda^3}{2 \cdot 3 \cdot \mu^3} \cdot \pi_0$$

⋮

$$\pi_i = \frac{\lambda^i}{i! \mu^i} \cdot \pi_0$$

$$\sum_{n=0}^K \pi_n = 1 \quad \sum_{n=0}^K \pi_0 \left(\frac{\lambda}{\mu} \right)^n / n! = 1$$

$$\pi_0 = \frac{1}{\sum_{n=0}^K \left(\frac{\lambda}{\mu} \right)^n / n!}$$

$$\pi_i = \frac{\left(\lambda / \mu \right)^i / i!}{\sum_{n=0}^K \left(\lambda / \mu \right)^n / n!}$$

M/M/k/k formulas

- Limiting probability of i customers in system:

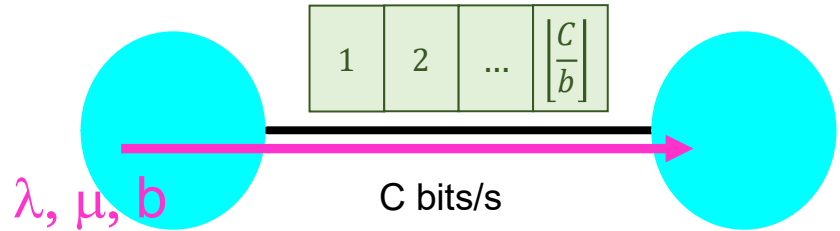
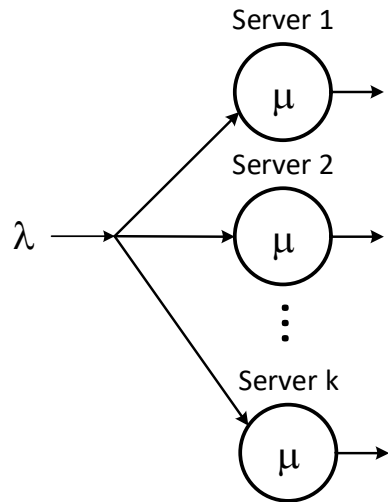
$$\pi_i = \frac{(\lambda/\mu)^i / i!}{\sum_{n=0}^k (\lambda/\mu)^n / n!}$$

- Probability that one arrival finds all k servers busy – blocking probability (Erlang B formula):

$$P_B = \pi_k = \frac{(\lambda/\mu)^k / k!}{\sum_{n=0}^k (\lambda/\mu)^n / n!}$$

- **Insensitivity property:** the Erlang B formula holds for any service distribution, not just exponentially distributed!

Point-to-point links and M/M/k/k



Erlang B formula:

$$P_B = \frac{(\lambda/\mu)^k / k!}{\sum_{n=0}^k (\lambda/\mu)^n / n!}$$

Point-to-point link = M/M/k/k:

- ❖ Circuit switching
- ❖ Call arrivals according to Poisson process at rate λ
- ❖ Exponentially distributed call durations with mean $1/\mu$
- ❖ Independence between arrivals and durations
- ❖ Each call occupies a bandwidth b
- ❖ Number of available circuits $k = \lfloor C/b \rfloor$

Point-to-point links and M/M/m/m

- A fixed phone operator wants to establish a direct link between Aveiro and Lisboa. Calls are offered according to a Poisson process at a rate of 30 calls/minute. The duration of calls is independent and exponentially distributed with mean 3 minutes. The duration of calls and its inter-arrival times are independent. The operator must provide at least 107 circuits to assure a blocking probability lower than 0.01.

$$P_B = \frac{(\lambda/\mu)^k/k!}{\sum_{n=0}^k (\lambda/\mu)^n/n!}$$

$\lambda = 30$
 $1/\mu = 3$
 $\frac{\lambda}{\mu} = 90 \text{ Erlangs}$

$$\frac{90^k/k!}{\sum_{n=0}^k 90^n/n!} \leq 0.01 \Rightarrow K = 107$$

M/M/k/k capacity provisioning

- What is the number of servers required to achieve a blocking probability less than α ?
- Example: $\alpha = 0.01$ (1% blocking), $\mu = 1$

$$P_B = \frac{(\lambda/\mu)^k / k!}{\sum_{n=0}^k (\lambda/\mu)^n / n!}$$

λ	k
$\lambda = 1$	5
$\lambda = 2$	7
$\lambda = 4$	10
$\lambda = 8$	15

Doubling λ does not double the number of required servers!

M/G/1 system

- Queuing system where:
 - There is one server
 - Customers arrive according to a Poisson process with rate λ
 - The service time has a general distribution
 - The arrival process is independent of the service time
 - The system accommodates an infinite number of customers
- Can not be modeled by a birth-death process.
- Pollaczek–Khinchine formula:

$$W_Q = \frac{\lambda E(S^2)}{2(1 - \lambda E(S))}$$

M/G/1 system

$$W_Q = \frac{\lambda E(S^2)}{2(1 - \lambda E(S))}$$

- When the service time is exponentially distributed, the M/G/1 reduces to M/M/1. Since $E(S) = 1/\mu$ and $E(S^2) = 2/\mu^2$

$$W_Q = \frac{\lambda}{\mu(\mu - \lambda)}$$

- When the service times are the same for all customers with value $1/\mu$, the M/G/1 reduces to M/D/1. Since $E(S) = 1/\mu$ and $E(S^2) = 1/\mu^2$

$$W_Q = \frac{\lambda}{2\mu(\mu - \lambda)}$$

- NOTE: Less variability, less delay!

$$W_Q = \frac{\lambda E(S^2)}{2(1 - \lambda E(S))}$$

$$S \sim \text{Exp}(\mu)$$

$$E(S) = 1/\mu \quad \text{Var}(S) = 1/\mu^2$$

$$\begin{aligned} E(S^2) &= \text{Var}(S) + [E(S)]^2 \\ &= \frac{1}{\mu^2} + \frac{1}{\mu^2} = \frac{2}{\mu^2} \end{aligned}$$

$$W_Q = \frac{\lambda \cdot 2/\mu^2}{2(1 - \lambda/\mu)} = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$W_Q = \frac{\lambda E(S^2)}{2(1 - \lambda E(S))}$$

$$\rho = 1/\mu$$

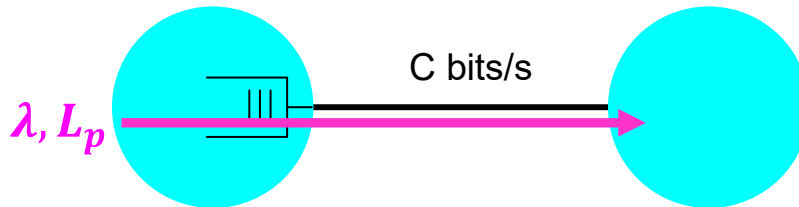
$$E(S) = 1/\mu \quad \text{Var}(S) = 0$$

$$E(S^2) = [E(S)]^2 = 1/\mu^2$$

$$W_Q = \frac{\lambda/\mu^2}{2(1 - \lambda/\mu)} = \frac{\lambda}{2\mu(\mu - \lambda)}$$

M/G/1 system

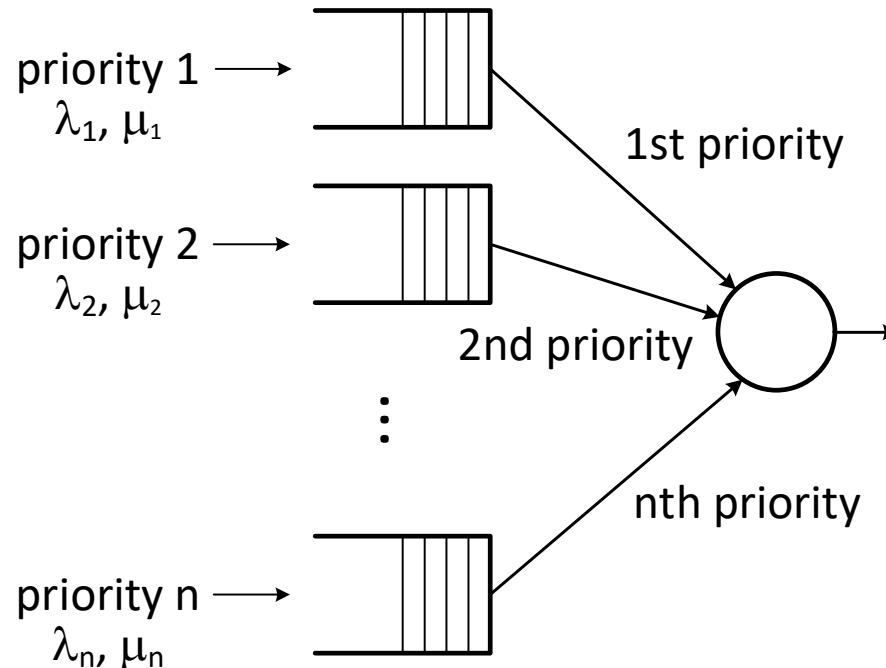
- A point-to-point link where packets arrive according to a Poisson process can be modelled by an M/G/1 system. If the length of packets is exponentially distributed it degenerates into an M/M/1 system. If the length of packets is fixed it degenerates into an M/D/1 system.



Point-to-point link = M/D/1:

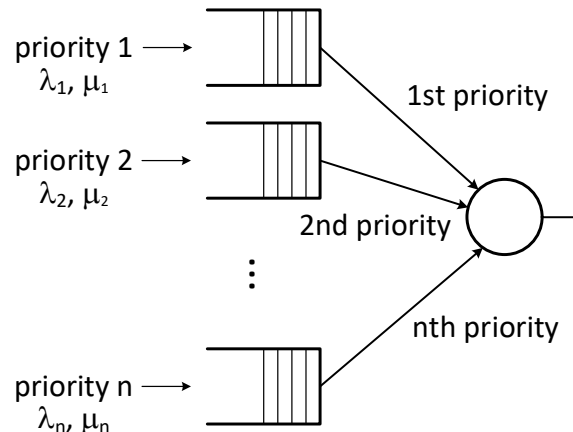
- ❖ Packet switching
- ❖ Poisson arrivals at rate λ
- ❖ Fixed packet lengths
 $\Rightarrow$ Service rate $\mu = C/L_p$
- ❖ Independence of arrivals and lengths
- ❖ Infinite capacity buffer

M/G/1 system with strict priorities



M/G/1 system with strict priorities

- There are n service classes. Lower index classes have higher ~~the~~ priority. Within each class clients are served with FCFS order. Service is not interrupted by arrival of higher priority client = **non-preemptive** service discipline.
- Class k is determined by arrival rate, λ_k , first and second moments of the service time, $E(S_k) = 1/\mu_k$, $E(S_k^2)$, and priority. Arrival process independent of service times in each class.

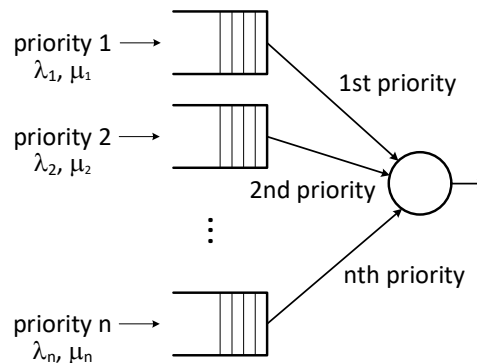


M/G/1 system with strict priorities

- The average delay in queue of class k is given by

$$W_{Qk} = \frac{\sum_{i=1}^n \lambda_i E(S_i^2)}{2(1 - \sum_{i=1}^k \rho_i)(1 - \sum_{i=1}^{k-1} \rho_i)}$$

where $\rho_k = \lambda_k / \mu_k$ and $\rho = \rho_1 + \dots + \rho_n < 1$.



M/G/1 system with strict priorities

- If the service times of each class are exponentially distributed with mean $1/\mu$ (non-preemptive M/M/1 system with priorities):

$$W_{Qk} = \frac{\rho/\mu}{(1 - \sum_{i=1}^k \rho_i)(1 - \sum_{i=1}^{k-1} \rho_i)}$$

- If the service times are fixed and equal to $1/\mu$ (non-preemptive M/D/1 system with priorities):

$$W_{Qk} = \frac{\rho/2\mu}{(1 - \sum_{i=1}^k \rho_i)(1 - \sum_{i=1}^{k-1} \rho_i)}$$

M/G/1 system with strict priorities

- **Give priority to smaller jobs:** Suppose there are two classes of clients – S (small) and L (large) – with arrival rates λ_S and λ_L and mean service sizes $E(S_S)$ and $E(S_L)$ where $E(S_S) < E(S_L)$. It can be shown that in order to minimize the average queuing delay over the two classes

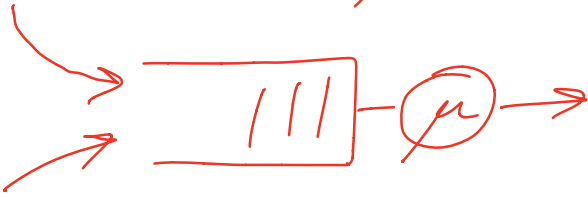
$$W_Q = \frac{\lambda_S W_Q^S + \lambda_L W_Q^L}{\lambda_S + \lambda_L}$$

class S should be given priority over class L.

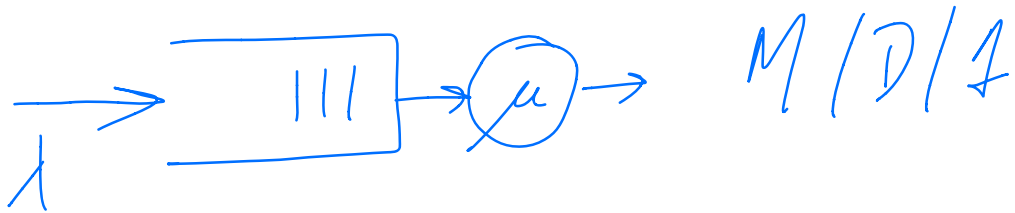
M/G/1 example

- Consider a point-to-point link shared by two packet flows. Within each flow packets are generated with a fixed length of 1000 bits and according to a Poisson process with rate 400 packets/second. The point-to-point link has 1 Mb/s of capacity. One of the flows must have an average packet delay lower than 2 ms and the other lower than 5 ms.
 - Verify that this specification can not be met if both flows are mixed in a single queue and served according to a FIFO discipline.
 - Is (strict) priority a solution to this problem?

$$\lambda_1 = 400 \text{ pkts/s}, L_{p1} = 1000 \text{ bits}$$



$$\lambda_2 = 400 \text{ pkts/s}, L_{p2} = 1000 \text{ bits}$$



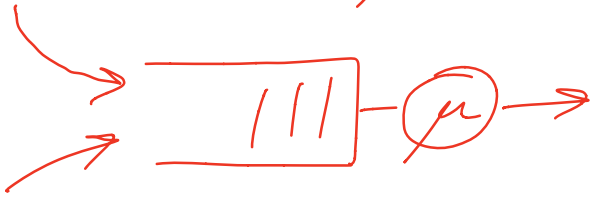
$$\lambda = \lambda_1 + \lambda_2 = 800 \text{ pkts/s}$$

$$\mu = \frac{C}{L} = \frac{1 \text{ M}}{1 \text{ K}} = 1000 \text{ pkts/s}$$

$$\rho = \frac{\lambda}{\mu} = 0.8$$

$$W = \frac{\rho / 2 \mu}{1 - \rho} + \frac{1}{\mu} = 3 \text{ ms}$$

$\lambda_1 = 400 \text{ pkts/s}$, $L_{p1} = 1000 \text{ bits}$, priority 1



$\lambda_2 = 400 \text{ pkts/s}$, $L_{p2} = 1000 \text{ bits}$, priority 2

non-preemptive M/D/1 with
strict priorities

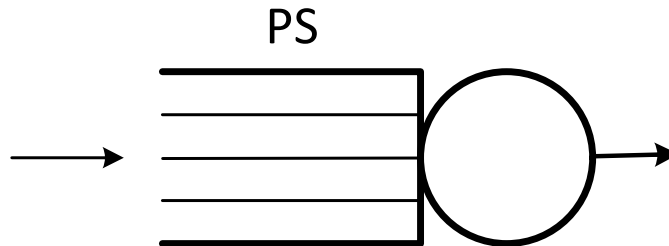
$$\rho_1 = \frac{\lambda_1}{\mu} = 0.4 \quad \rho_2 = \frac{\lambda_2}{\mu} = 0.4 \quad \rho = \rho_1 + \rho_2 = 0.8$$

$$W_1 = \frac{\rho/2\mu}{1-\rho_1} + \frac{1}{\mu} = 1.7 \text{ ms} < 2 \text{ ms}$$

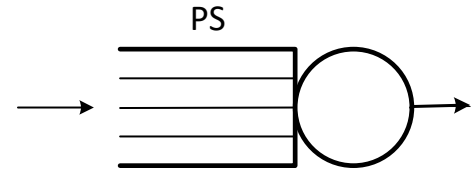
$$W_2 = \frac{\rho/2\mu}{(1-\rho_1)(1-\rho_1-\rho_2)} + \frac{1}{\mu} = 4.3 \text{ ms} < 5 \text{ ms}$$

Processor-Sharing scheduling

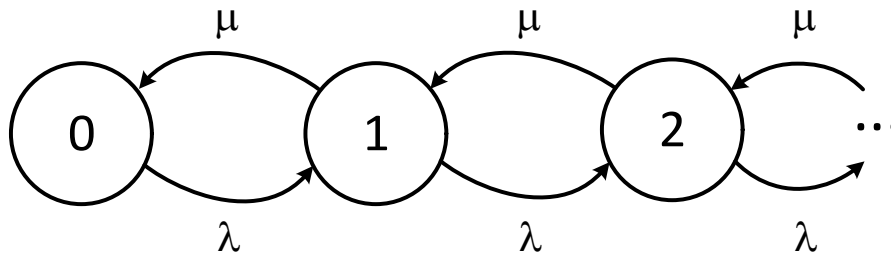
- **Processor-Sharing (PS) scheduling** – every client in the queue receives service at all times; if the server has service rate μ and there are n clients at the server, each client is served at rate μ/n



Processor-Sharing scheduling



- $M/M/1/PS = M/M/1$ from the performance point-of-view
- $M/G/1/PS = M/M/1/PS$
 - Limiting probabilities independent of service time distribution – **insensitivity property**



$$W = \frac{1}{\mu - \lambda}$$

$$L = \frac{\lambda}{\mu - \lambda}$$

If there are n in system, each is served with rate μ/n , and death is at rate μ